1. What is the business objective of a flight price prediction system?

The primary business objective is to **forecast future flight ticket prices accurately** based on historical data and influencing factors such as route, airline, travel date, and time. This enables **cost optimization, demand planning, and informed decision-making** for both customers and airlines, while improving pricing transparency and competitiveness.

---

2. Is flight price prediction a regression or classification problem? Why?

Flight price prediction is a **regression problem** because the target variable—**ticket price—is a continuous numerical value**. The goal is to predict an exact or approximate price rather than assigning the flight to a predefined category or class.

---

3. Who are the primary stakeholders for this model?

Primary stakeholders include:

- **Customers / Travelers** – to identify optimal booking times and reduce travel costs.
- **Airlines** – for dynamic pricing, revenue management, and demand forecasting.
- **Travel Agencies / Booking Platforms** – to offer price insights and recommendations.
- **Business Travelers & Corporates** – for travel budgeting and expense forecasting.

---

4. How would airlines vs customers use your prediction results differently?

- **Airlines** use predictions to:
    - Optimize dynamic pricing strategies
    - Maximize revenue based on demand and seasonality
    - Adjust seat inventory and promotional offers
- **Customers** use predictions to:
    - Decide the best time to book tickets
    - Compare routes and airlines for cost efficiency
    - Plan travel budgets in advance

Thus, airlines use predictions for **revenue optimization**, while customers use them for **cost savings**.

---

5. What business risks arise from inaccurate price predictions?

Inaccurate predictions can lead to:

- **Revenue loss** for airlines due to underpricing or overpricing
- **Loss of customer trust** if predicted prices differ significantly from actual prices

- **Poor booking decisions**, impacting customer satisfaction
- **Operational inefficiencies** in inventory and demand planning
  Overall, unreliable predictions can negatively affect both **profitability and brand credibility**.

---

6. How would you define success metrics for this project from a business perspective?

From a business perspective, success can be measured using:

- **Low prediction error** (RMSE or MAE) indicating pricing accuracy
- **High customer engagement** or increased booking conversions
- **Reduction in customer complaints related to pricing**
- **Improved revenue or margin stability** for airlines
- **Consistent performance during peak seasons**

A successful model delivers **accurate, stable predictions that support business decisions and enhance user trust**.

---

1. What features typically influence flight prices?

Flight prices are influenced by multiple factors, including:

- **Route (source–destination)**
- **Airline type** (full-service vs low-cost)
- **Date of journey** (seasonality, holidays)
- **Departure and arrival time**
- **Number of stops**
- **Journey duration**
- **Booking window** (days before departure)

These features reflect **demand, operational cost, and customer willingness to pay**.

---

2. How does departure time affect flight pricing?

Departure time affects pricing due to **demand and convenience**:

- **Early morning and late-night flights** are usually cheaper due to lower demand.
- **Peak hours (morning and evening)** are priced higher because they align with business and leisure travel schedules.
- Flights departing at convenient times command a **premium price**.

---

3. Why are weekends and holidays important features?

Weekends and holidays drive **higher travel demand**, which directly impacts pricing:

- Leisure travelers typically book flights on weekends and during festivals.
- Airlines increase prices due to **demand surge and limited seat availability**.
- Ignoring these features can lead to **systematic underestimation** of prices.

---

4. How do you analyze seasonality in flight prices?

Seasonality can be analyzed by:

- Extracting **month, week, and day** from the travel date
- Plotting **average price trends by month or quarter**
- Comparing **peak vs off-season pricing**
- Using time-series visualizations to detect recurring patterns

This helps identify **cyclical trends** such as festive spikes or off-season dips.

---

5. What patterns did you observe during EDA?

Typical EDA observations include:

- Prices increase during **holiday months and weekends**
- **Non-stop flights** are generally more expensive than connecting flights
- Certain airlines consistently price higher due to service quality
- Longer journey durations often correlate with **lower prices**
- Some routes show persistent premium pricing due to demand imbalance

---

6. How do airline type and route affect pricing?

- **Airline type**:
    - Full-service airlines charge higher prices due to added amenities
    - Budget airlines offer lower base fares but fewer services
- **Route**:
    - High-demand or business-heavy routes are more expensive
    - Routes with limited airline competition have higher prices

Thus, pricing reflects both **brand positioning and market competition**.

---

7. How would you identify outliers in flight price data?

Outliers can be identified using:

- **Box plots** to detect extreme values
- **IQR method** to flag unusually high or low prices
- **Z-score analysis** for statistical deviation
- Visual inspection through **scatter plots**

Outliers are evaluated carefully, as extreme prices may represent **valid premium or discounted tickets**.

---

8. How do you check multicollinearity in features?

Multicollinearity is checked by:

- Calculating the **correlation matrix**
- Computing **Variance Inflation Factor (VIF)** values
- Observing instability in regression coefficients

High multicollinearity can reduce model interpretability and is addressed through **feature selection or dimensionality reduction**.

---

3. Data Cleaning & Preprocessing

How did you handle missing values in your dataset?

Missing values were handled based on feature importance:

- Numerical features were imputed using **median** to reduce outlier impact
- Categorical features were imputed using **mode** or labeled as "Unknown"
- Rows with excessive missing critical values were dropped

Why is date-time feature extraction important?

Date-time columns contain **hidden predictive signals** like seasonality, weekday/weekend effects, and booking patterns. Extracting these improves model accuracy and interpretability.

How did you convert categorical variables into numerical form?

Categorical variables were converted using **encoding techniques** so that machine learning models could process them numerically.

What encoding technique did you use and why?

- **One-Hot Encoding** for low-cardinality categorical features (Airline, Source, Destination)
- It avoids introducing ordinal relationships and works well with tree-based models

---

How do you handle inconsistent airline or city names?

- Standardized text using **lowercasing, trimming spaces**
- Corrected spelling inconsistencies
- Used mapping dictionaries to merge duplicate categories

How do you scale numerical features and why?

- Applied **StandardScaler** or **MinMaxScaler**
- Scaling ensures features contribute equally, especially for distance-based or linear models

---

4. Feature Engineering (Very Important)

Which new features did you create from the Date_of_Journey column?

- Journey **day**
- Journey **month**
- **Weekday / Weekend**
- **Season / Peak month indicator**

How did you extract information from Dep_Time and Arrival_Time?

- Extracted **hour and minute**
- Categorized into **Morning / Afternoon / Evening / Night** slots

---

Why is journey duration a critical feature?

Journey duration captures **convenience, layovers, and operational cost**, which directly influence ticket pricing.

How did you calculate total travel time?

By converting duration strings into minutes and summing **hours + minutes** into a single numerical feature.

How did you handle stops (Total_Stops) feature?

- Converted textual categories (non-stop, 1 stop) into numerical values
- This preserves the ordinal nature of stops

How do holidays or peak seasons affect pricing, and how would you encode them?

- Prices spike during high-demand periods
- Encoded using:
    - Binary holiday flag
    - Seasonal indicators (peak vs off-season)

Which feature had the highest correlation with price?

Typically:

- **Total_Stops**
- **Airline**
- **Journey Duration**
- **Travel Month**

---

5. Machine Learning Model Selection

Why did you choose regression models for this problem?

Because the target variable—**flight price—is continuous**, making regression the appropriate modeling approach.

Which algorithms did you try and why?

- Linear Regression (baseline)
- Decision Tree Regressor
- Random Forest Regressor
- XGBoost Regressor
  To compare **bias–variance tradeoffs** and performance.

Why do tree-based models often perform better for price prediction?

They:

- Capture non-linear relationships

- Handle categorical encodings well
- Are robust to outliers and feature interactions

What are the drawbacks of linear regression in this problem?

- Assumes linearity
- Sensitive to outliers
- Performs poorly with complex interactions

Why did Random Forest/XGBoost outperform other models?

They leverage **ensemble learning**, capturing complex patterns while reducing variance and overfitting.

How do ensemble models reduce overfitting?

By combining predictions from multiple weak learners, averaging errors and improving generalization.

---

6. Model Training & Evaluation

Which evaluation metrics did you use and why?

- **RMSE**: Penalizes large errors
- **MAE**: Easy interpretability
- **R^2**: Explains variance captured

Why is RMSE preferred over MAE in price prediction?

RMSE penalizes large price deviations more heavily, which is critical in financial predictions.

What does R^2 score represent?

It represents the **percentage of variance in price explained by the model**.

How do you interpret a low RMSE but low R^2?

The model predicts average prices well but fails to explain variability across different routes or seasons.

How did you handle overfitting?

- Cross-validation
- Hyperparameter tuning
- Limiting tree depth and number of estimators

Why is cross-validation important?

It ensures **model stability and generalization** across unseen data.

How did hyperparameter tuning improve performance?

It optimized model complexity, balancing bias and variance for better accuracy.

---

7. Model Explainability & Interpretation

How do you explain predictions to non-technical stakeholders?

By explaining **key drivers** (route, stops, season) rather than algorithm details.

What is feature importance?

A measure of how much each feature contributes to the model's prediction.

How do SHAP values help in this project?

They explain **individual predictions**, showing how each feature increases or decreases price.

Which features had the highest impact on pricing?

- Number of stops
- Airline
- Travel month
- Journey duration

Why is explainability important in ML models?

It builds **trust, transparency, and regulatory compliance**.

---

8. MLflow & Experiment Tracking

What is MLflow and why did you use it?

MLflow is used to **track experiments, metrics, parameters, and models** systematically.

What components of MLflow did you use?

- MLflow Tracking
- Model Registry
- Artifact storage

How does MLflow help in model versioning?

Each model version is logged with metadata, enabling rollback and comparison.

How did you compare multiple experiments?

By comparing logged metrics like RMSE and R^2 across runs.

How do you register a model using MLflow?

By logging it to the **Model Registry** with a version and stage (Staging/Production).

---

9. Deployment Using Streamlit

Why did you choose Streamlit for deployment?

It enables **rapid, interactive ML app development** with minimal frontend effort.

How does your Streamlit app work end-to-end?

User inputs → preprocessing → model inference → price prediction displayed.

How do users interact with the model?

Through dropdowns, date pickers, and time selectors.

How did you handle invalid user inputs?

By setting input constraints and default fallback values.

How did you optimize performance in Streamlit?

- Cached models using @st.cache_resource
- Minimized recomputation

How do you load the trained model in the app?

Using joblib or MLflow model loading utilities.

---

10. Testing & Validation

How did you validate predictions after deployment?

By comparing predicted prices with real-world prices periodically.

How do you handle unseen categories during inference?

Mapped them to "Other" or default encoded values.

What happens if the user selects a route not present in training data?

The model falls back to generalized patterns with lower confidence.

How do you ensure reproducibility of results?

- Fixed random seeds
- Version-controlled data and models
- Logged experiments in MLflow

---

11. Advanced / Scenario-Based Questions

How would you retrain your model?

Scheduled retraining using recent data (weekly/monthly).

How would you handle real-time price prediction?

By deploying the model behind an API with real-time feature ingestion.

What external data could improve accuracy?

- Weather
- Holidays

- Fuel prices
- Demand signals

How would you detect data drift?

By monitoring feature distributions and prediction errors over time.

How would you scale this system?

Using cloud infrastructure, APIs, and distributed inference.

If the model fails during festive seasons, how would you fix it?

Add festive data, introduce holiday features, and retrain the model.

---

12. SQL & Data Engineering

```
SELECT source, destination, AVG(price)
FROM flights
GROUP BY source, destination;
SELECT source, destination, AVG(price) AS avg_price
FROM flights
GROUP BY source, destination
ORDER BY avg_price DESC;
SELECT *
FROM flights
WHERE price > (
  SELECT AVG(price) + 2*STDDEV(price) FROM flights
);
```

Schema: flight_id, airline, source, destination, date, price, stops, duration

---

13. Common Coding / Test Questions

```
df['month'] = df['date'].dt.month
df['day'] = df['date'].dt.day
pd.get_dummies(df, columns=['airline'])
```

```
from sklearn.ensemble import RandomForestRegressor

model = RandomForestRegressor()

from sklearn.metrics import mean_squared_error

rmse = mean_squared_error(y_true, y_pred, squared=False)

import joblib

joblib.dump(model, 'model.pkl')

model = joblib.load('model.pkl')
```

---

14. HR + Project Defense

Explain your project in 2 minutes

I built an end-to-end ML system to predict flight prices using historical data, performed EDA and feature engineering, trained regression models, tracked experiments with MLflow, and deployed the model using Streamlit for real-time user interaction.

What challenges did you face?

Handling categorical variables, seasonality, and overfitting.

What would you improve?

Add real-time data and advanced demand signals.

Why should we trust your model?

It is validated, explainable, and consistently monitored.

How is this relevant to our business?

It supports pricing optimization, customer decision-making, and revenue forecasting.

---

1. Machine Learning Fundamentals

1. What is Machine Learning?

Machine Learning is a subset of AI that enables systems to **learn patterns from data and make predictions or decisions without being explicitly programmed**.

---

2. What are the main types of Machine Learning?

- **Supervised Learning** – labeled data (regression, classification)
- **Unsupervised Learning** – unlabeled data (clustering, dimensionality reduction)
- **Semi-Supervised Learning** – limited labeled data
- **Reinforcement Learning** – learning via rewards and penalties

---

3. What is the difference between supervised and unsupervised learning?

Supervised learning uses **labeled data**, while unsupervised learning identifies **hidden patterns in unlabeled data**.

---

4. What is regression vs classification?

- **Regression** predicts continuous values
- **Classification** predicts categorical labels

---

2. Data Preprocessing & Feature Engineering

5. Why is data preprocessing important?

It improves **data quality, model performance, and training stability** by handling missing values, scaling, and encoding.

---

6. How do you handle missing values?

- Drop rows/columns
- Impute using mean, median, mode
- Use model-based imputation

---

7. What is feature scaling and why is it needed?

Scaling standardizes feature ranges, which is crucial for **distance-based and gradient-based models**.

---

8. Difference between StandardScaler and MinMaxScaler?

- **StandardScaler**: zero mean, unit variance
- **MinMaxScaler**: scales values between 0 and 1

---

9. What is feature engineering?

Creating new, meaningful features from raw data to **improve model performance**.

---

3. Model Training & Evaluation

10. What is overfitting and underfitting?

- **Overfitting**: model learns noise, poor generalization
- **Underfitting**: model is too simple to capture patterns

---

11. How do you prevent overfitting?

- Cross-validation
- Regularization
- Feature selection
- Ensemble models

---

12. What is bias-variance tradeoff?

Balancing **model complexity and generalization** to minimize total error.

---

13. What is cross-validation?

A technique to evaluate models by splitting data into **multiple train-test folds**.

---

14. What evaluation metrics are used for regression?

- MAE
- RMSE
- R^2 score

---

15. Difference between RMSE and MAE?

RMSE penalizes **large errors more strongly** than MAE.

---

4. Popular Machine Learning Algorithms

16. Explain Linear Regression.

A model that predicts a continuous output by fitting a linear relationship between features and target.

---

17. What are the assumptions of Linear Regression?

- Linearity
- Independence
- Homoscedasticity
- Normality of errors
- No multicollinearity

---

18. What is Logistic Regression?

A classification algorithm that predicts **probabilities using a sigmoid function**.

---

19. What is KNN?

A non-parametric algorithm that classifies based on **nearest neighbors**.

---

20. What is Decision Tree?

A tree-based model that splits data using **feature-based rules**.

---

21. What is Random Forest?

An ensemble of multiple decision trees that reduces variance and improves accuracy.

---

22. What is Gradient Boosting?

A technique where models are trained **sequentially to correct previous errors**.

---

23. Difference between Random Forest and XGBoost?

Random Forest	**XGBoost**
Parallel trees	Sequential trees
Reduces variance	Reduces bias + variance
Less tuning	Highly optimized

---

5. Python for Machine Learning

24. Why is Python popular for ML?

- Simple syntax
- Rich ML libraries
- Strong community support

---

25. Difference between list and numpy array?

NumPy arrays are **faster and support vectorized operations**.

---

26. What is Pandas used for?

Data manipulation, cleaning, and analysis.

---

27. What is NumPy?

A library for **numerical computing and array operations**.

---

28. What is Scikit-learn?

A Python library providing **ML algorithms and utilities**.

---

6. Coding-Based Questions

29. Train a Linear Regression model

```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
```

---

30. Perform train-test split

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

---

31. Calculate RMSE

```
from sklearn.metrics import mean_squared_error
rmse = mean_squared_error(y_test, y_pred, squared=False)
```

---

32. Apply Standard Scaling

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

X_scaled = scaler.fit_transform(X)
```

---

7. Model Explainability

33. What is feature importance?

A measure of how much each feature contributes to predictions.

---

34. What is SHAP?

A method that explains **individual predictions** using game theory.

---

35. Why is explainability important?

It builds **trust, transparency, and regulatory compliance**.

---

8. Deployment & Production

36. What is model deployment?

Making a trained model available for **real-world use**.

---

37. How do you monitor model performance?

By tracking **prediction errors, data drift, and retraining needs**.

---

38. What is data drift?

Change in data distribution over time.

---

39. What is model versioning?

Tracking and managing multiple model versions.

---

9. Scenario-Based Questions

40. How would you improve a poorly performing model?

- Better features
- More data

- Hyperparameter tuning
- Algorithm change

---

41. What would you do if the dataset is imbalanced?

- Resampling
- Class weights
- SMOTE

---

42. How do you choose the right algorithm?

Based on:

- Data size
- Problem type
- Interpretability
- Performance

---

10. HR + Conceptual Wrap-Up

43. Explain ML to a non-technical person.

Machine Learning is teaching computers to **learn from past data and make smarter decisions automatically**.

---

44. What ML project are you most proud of?

Explain business problem, data, model, impact.

---

45. What are your strengths in ML?

Data understanding, feature engineering, model evaluation.